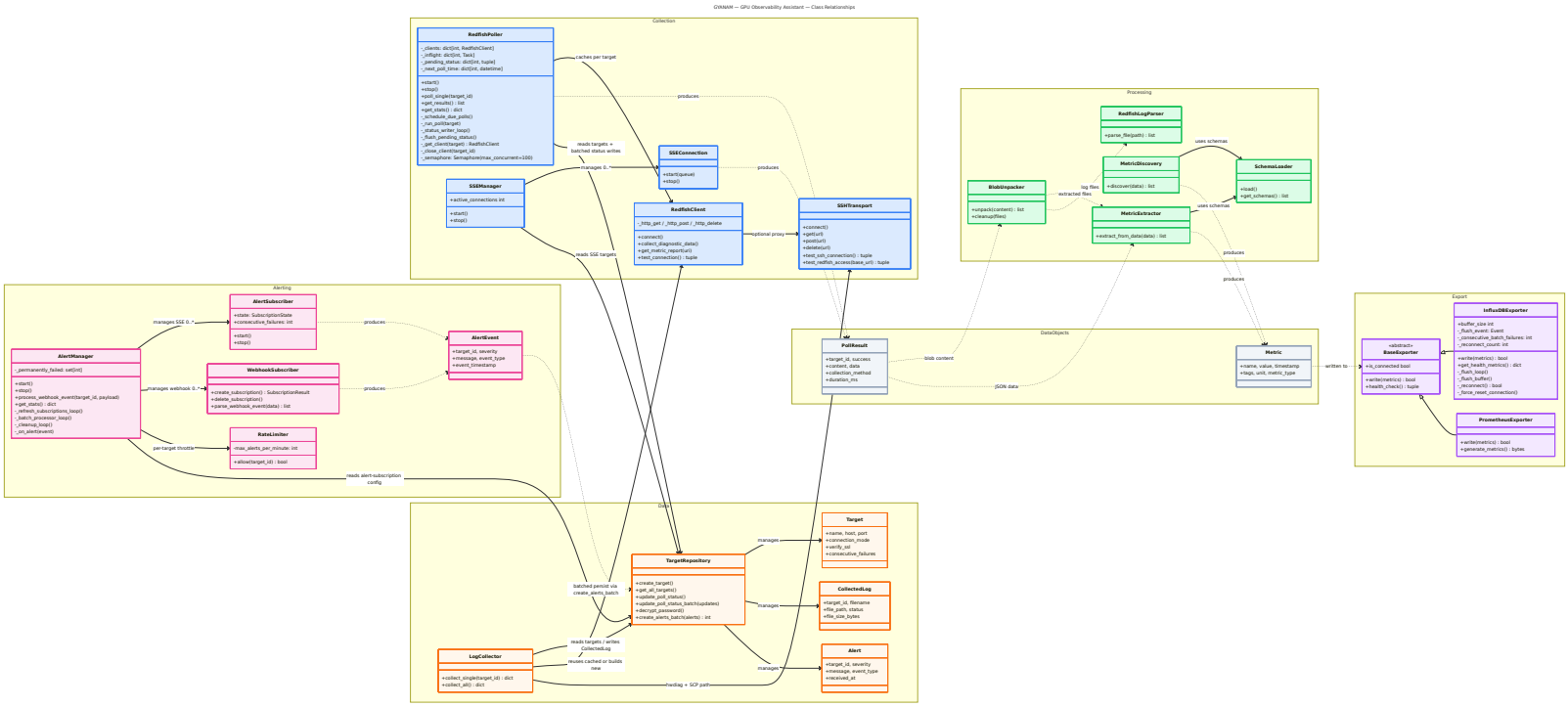
GVAMAM — GPU Observability Assistant — Class Relationships
Collection
RedfishPoller
_clients: dict[int, RedfishClient]
_inflight: dict[int, Task]
_pending_status: dict[int, tuple]
_next_poll_time: dict[int, datetime]
+start()
+stop()
+poll_single(target_id)
#get_results() : list
+get_stats() : dict
_schedule_due_polls()
_run_poll(target)
_status_writer_loop()
_flush_pending_status()
_get_client(target) : RedfishClient
_close_client(target_id)
_semaphore: Semaphore(max_concurrent=100)
caches per target
produces
reads targets + batched status writes
SSEConnection
+start(queue)
+stop()
produces
SSEManager
+active_connections int
+start()
+stop()
manages 0..*
RedfishClient
_http_get / _http_post / _http_delete
+connect()
+collect_diagnostic_data()
+get_metric_report(url)
+test_connection() : tuple
optional proxy
SSHTransport
+connect()
+get(url)
+post(url)
+delete(url)
+test_sub_connection() : tuple
+test_redfish_access(base_url) : tuple
reads SSE targets
Processing
RedfishLogParser
+parse_file(path) : list
MetricDiscovery
+discover(data) : list
uses schemas
SchemaLoader
+load()
+get_schemas() : list
BlobUnpacker
+unpack(content) : list
+cleanup(files)
log files
extracted files
MetricExtractor
+extract_from_data(data) : list
uses schemas
produces
produces
Alerting
AlertSubscriber
+state: SubscriptionState
+consecutive_failures: int
+start()
+stop()
manages SSE 0..*
produces
AlertManager
_permanently_failed: set[int]
+start()
+stop()
+process_webhook_event(target_id, payload)
+get_stats() : dict
_refresh_subscriptions_loop()
_batch_processor_loop()
_cleanup_loop()
_on_alert(event)
manages webhook 0..*
WebhookSubscriber
+create_subscription() : SubscriptionResult
+delete_subscription()
+parse_webhook_event(data) : list
produces
AlertEvent
+target_id, severity
+message, event_type
+event_timestamp
RateLimiter
-max_alerts_per_minute: int
+allow(target_id) : bool
per-target throttle
reads alert-subscription config
DataObjects
PollResult
+target_id, success
+content_data
+collection_method
+duration_ms
blob content
JSON data
Metric
+name, value, timestamp
+tags, unit, metric_type
written to
Export
<<abstract>>
BaseExporter
+is_connected bool
+write(metrics) : bool
+health_check() : tuple
InfluxDBExporter
+buffer_size int
- _flush_event: Event
- _consecutive_batch_failures: int
- _reconnect_count: int
+write(metrics) : bool
+get_health_metrics() : dict
- _flush_loop()
- _flush_buffer()
- _reconnect() : bool
- _force_reset_connection()
PrometheusExporter
+write(metrics) : bool
+generate_metrics() : bytes
Data
Target
+name, host, port
+connection_mode
+verify_ssl
+consecutive_failures
TargetRepository
+create_target()
+get_all_targets()
+update_poll_status()
+update_poll_status_batch(updates)
+decrypt_password()
+create_alerts_batch(alerts) : int
manages
batched persist via create_alerts_batch
CollectedLog
+target_id, filename
+file_path, status
+file_size_bytes
manages
Alert
+target_id, severity
+message, event_type
+received_at
manages
LogCollector
+collect_single(target_id) : dict
+collect_all() : dict
reads targets / writes CollectedLog
reuses cached or builds new
rsyslog + SCP path